Deep Generative Models

Chapter 1: Text Generation via Language Models

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Transformer: *Revolutionary Sequence Models*

Transformer was proposed in 2017 in the paper

- Attention is All You Need!

providing a computationally-efficient way for sequence processing

Key Component of Transformers

Transformers use the Attention mechanism which enables parallel processing of sequences[1]

We use the Attention mechanism in the sequel to build context!

[1] Read more in Chapter 7 of *Applied Deep Learning* lecture notes

Key-Query-Value Tuple

Say we have *the sequence of tokens*

$$x_1, x_2, \ldots, x_t$$

and intend to build context $\mathbf{c}_t$ *to find the distribution of next token* x_{t+1}

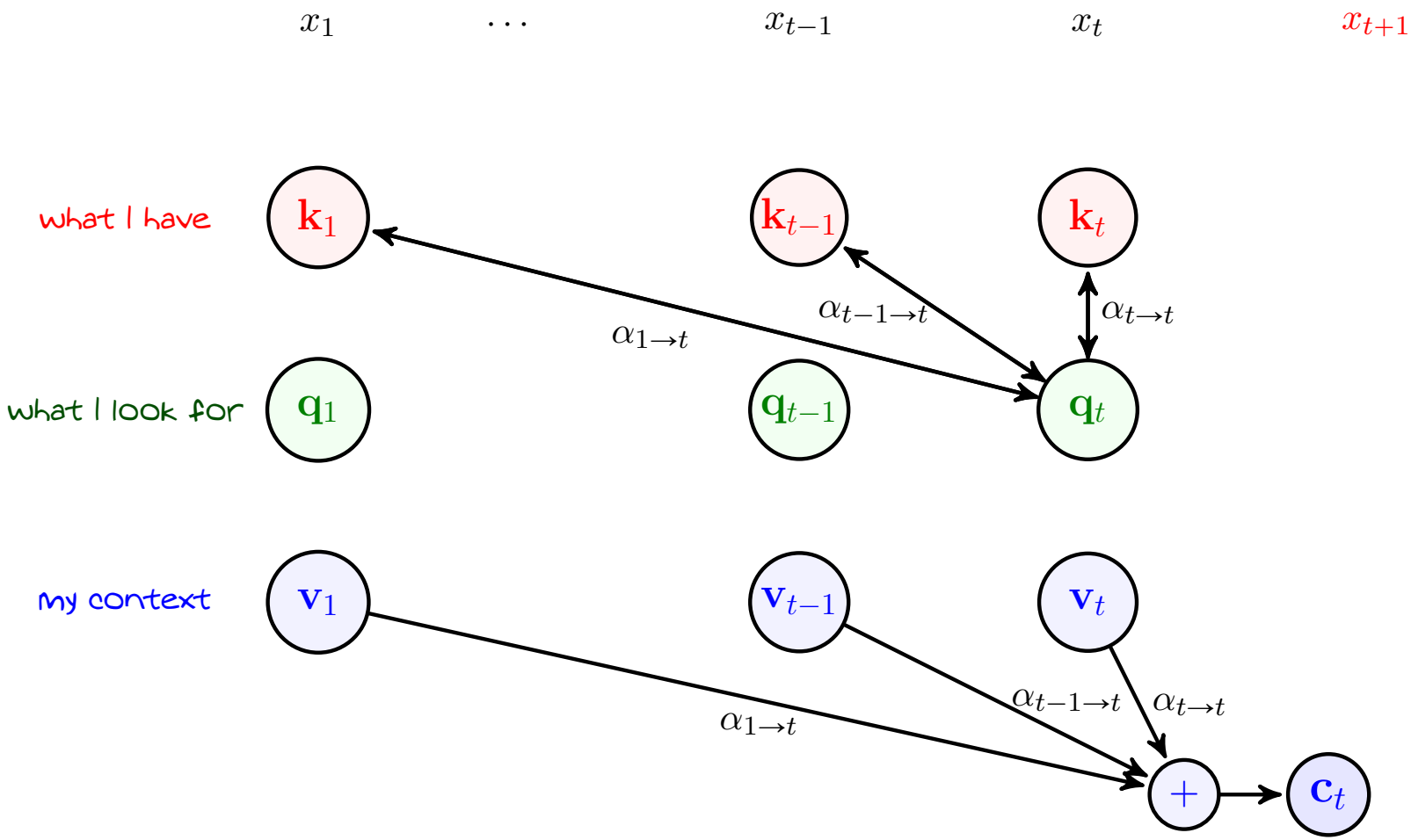
For each token in the sequence, we give three separate embeddings

- *Key* $\mathbf{k}_t \in \mathbb{R}^S$ *which flags what kind of context token* x_t *has to present*
- *Query* $\mathbf{q}_t \in \mathbb{R}^S$ *which flags what kind of context token* x_t *is looking for*
- *Value* $\mathbf{v}_t \in \mathbb{R}^S$ *which embeds the context of token* x_t

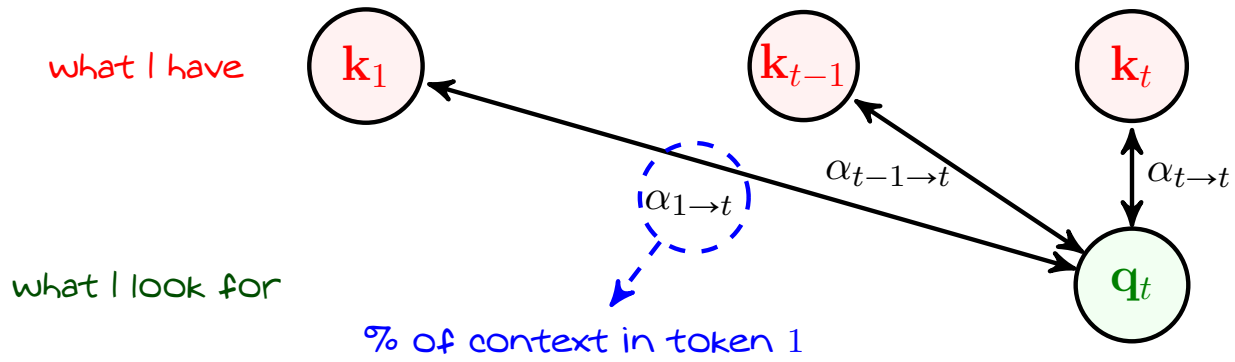
\+ *How do we compute these embeddings?*

– No worries! We'll see shortly!

Building Context by Attention



Computing Attention Weights



We can compare the query with each key using *inner product*

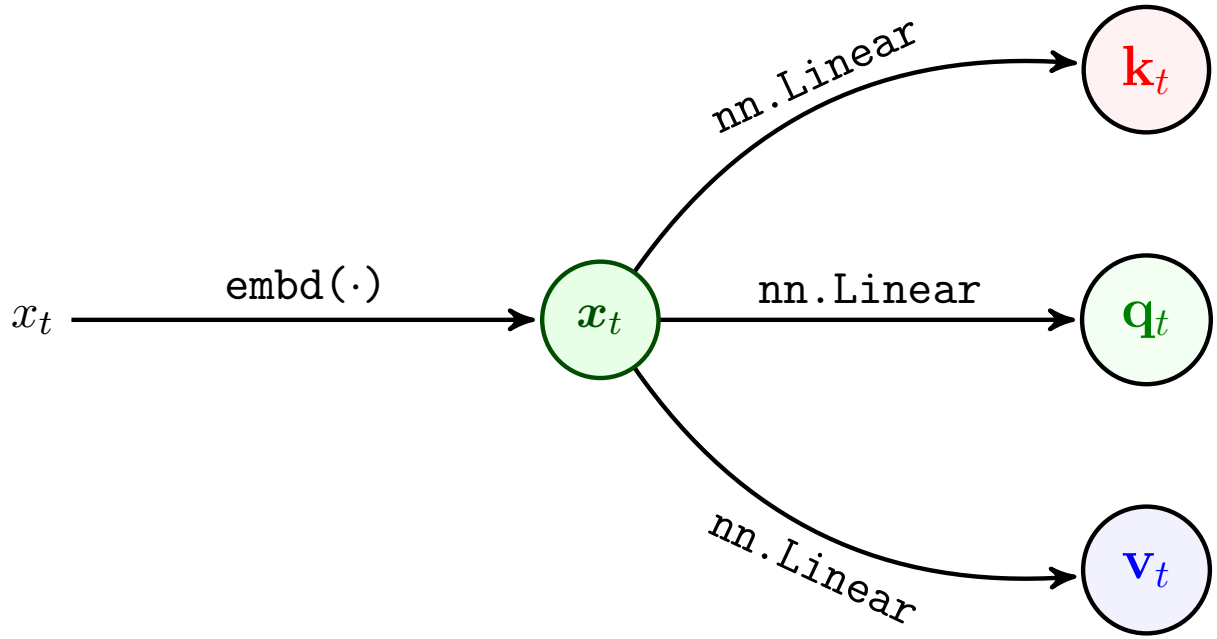
$$\xi_{j\to t} = \langle \mathbf{q}_t; \mathbf{k}_j \rangle$$

The larger $\xi_{j\to t}$ is, the more related tokens j and t are!

But, we need weights $\in [0, 1]$; so, we use Softmax

$$[\alpha_{j\to t} \text{ for } j = 1:t] = \text{Soft}_{\max}\left(\langle \mathbf{q}_t; \mathbf{k}_j \rangle \text{ for } j = 1:t\right)$$

Embedding Tokens: *Key, Query and Value*



We typically compute the tuples by *linear projections of token embedding*

$$\mathbf{k}_t = \mathbf{W}_k \boldsymbol{x}_t \qquad \mathbf{q}_t = \mathbf{W}_q \boldsymbol{x}_t \qquad \mathbf{v}_t = \mathbf{W}_v \boldsymbol{x}_t$$

fore some $\mathbf{H}_k, \mathbf{W}_q, \mathbf{W}_v \in \mathbb{R}^{S \times E}$

Sequential Order

Say we have the following two sequences

$$x_{1:t} = x_1, x_2, x_3 \ldots, x_t$$
$$\hat{x}_{1:t} = x_2, x_1, x_3 \ldots, x_t$$

We have $\mathbf{k}_i = \mathbf{W}_k \boldsymbol{x}_j$*; thus, the keys see the same permutation*

$$\hat{\mathbf{k}}_{1:t} = \mathbf{k}_2, \mathbf{k}_1, \mathbf{k}_3 \ldots, \mathbf{k}_t$$

If we compare with the query $\mathbf{q}_t$*, we get same permutation in weights*

$$\hat{\alpha}_{1\to t}, \hat{\alpha}_{2\to t}, \hat{\alpha}_{3\to t} \ldots, \hat{\alpha}_{t\to t} = \alpha_{2\to t}, \alpha_{1\to t}, \alpha_{3\to t} \ldots, \alpha_{t\to t}$$

Sequential Order

Weights observe the same permutation as input

$$\hat{\alpha}_{1\to t}, \hat{\alpha}_{2\to t}, \hat{\alpha}_{3\to t} \ldots, \hat{\alpha}_{t\to t} = \alpha_{2\to t}, \alpha_{1\to t}, \alpha_{3\to t} \ldots, \alpha_{t\to t}$$

We have further $\mathbf{v}_i = \mathbf{W}_v \boldsymbol{x}_j$*; thus, the values also see the same permutation*

$$\hat{\mathbf{v}}_{1:t} = \mathbf{v}_2, \mathbf{v}_1, \mathbf{v}_3 \ldots, \mathbf{v}_t$$

So, the context of permuted sequence is

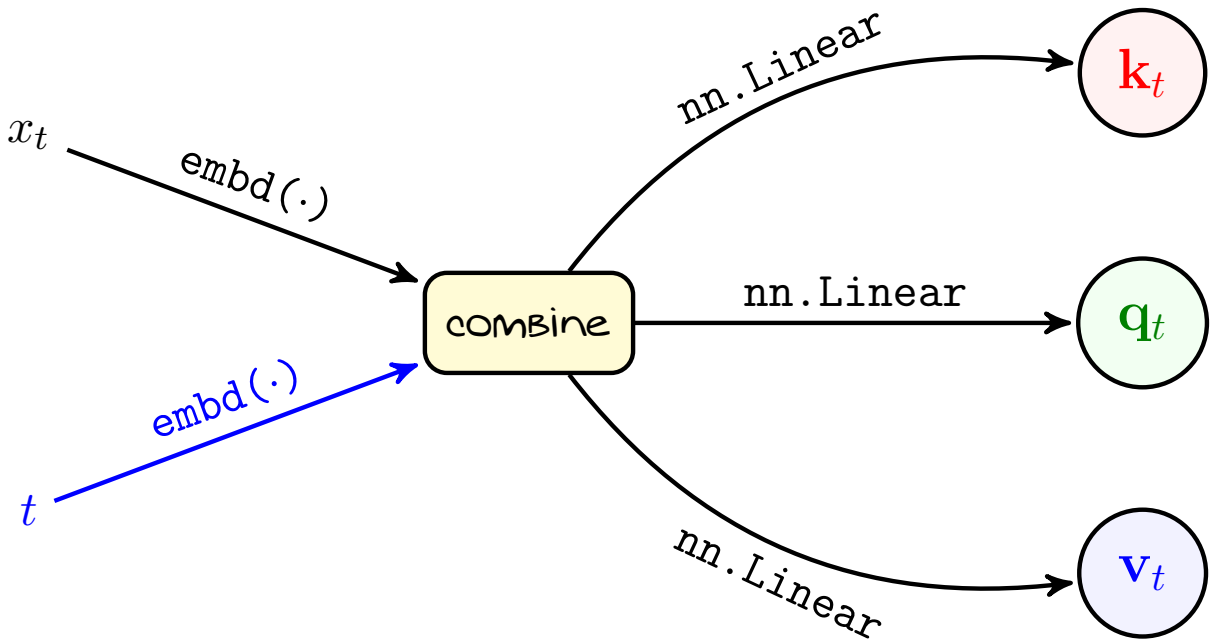
$$\begin{aligned}\hat{\mathbf{c}}_t &= \hat{\alpha}_{1\to t}\mathbf{v}_2 + \hat{\alpha}_{2\to t}\mathbf{v}_1 + \hat{\alpha}_{3\to t}\mathbf{v}_3 + \ldots + \hat{\alpha}_{t\to t}\mathbf{v}_t \\ &= \alpha_{2\to t}\mathbf{v}_2 + \alpha_{1\to t}\mathbf{v}_1 + \alpha_{3\to t}\mathbf{v}_3 + \ldots + \alpha_{t\to t}\mathbf{v}_t = \mathbf{c}_t\end{aligned}$$

Moral of Story

Using only token embedding we do not capture sequential order via attention!

Encoding Tokens *with Sequential Order*

We can capture sequential order by *positional encoding*

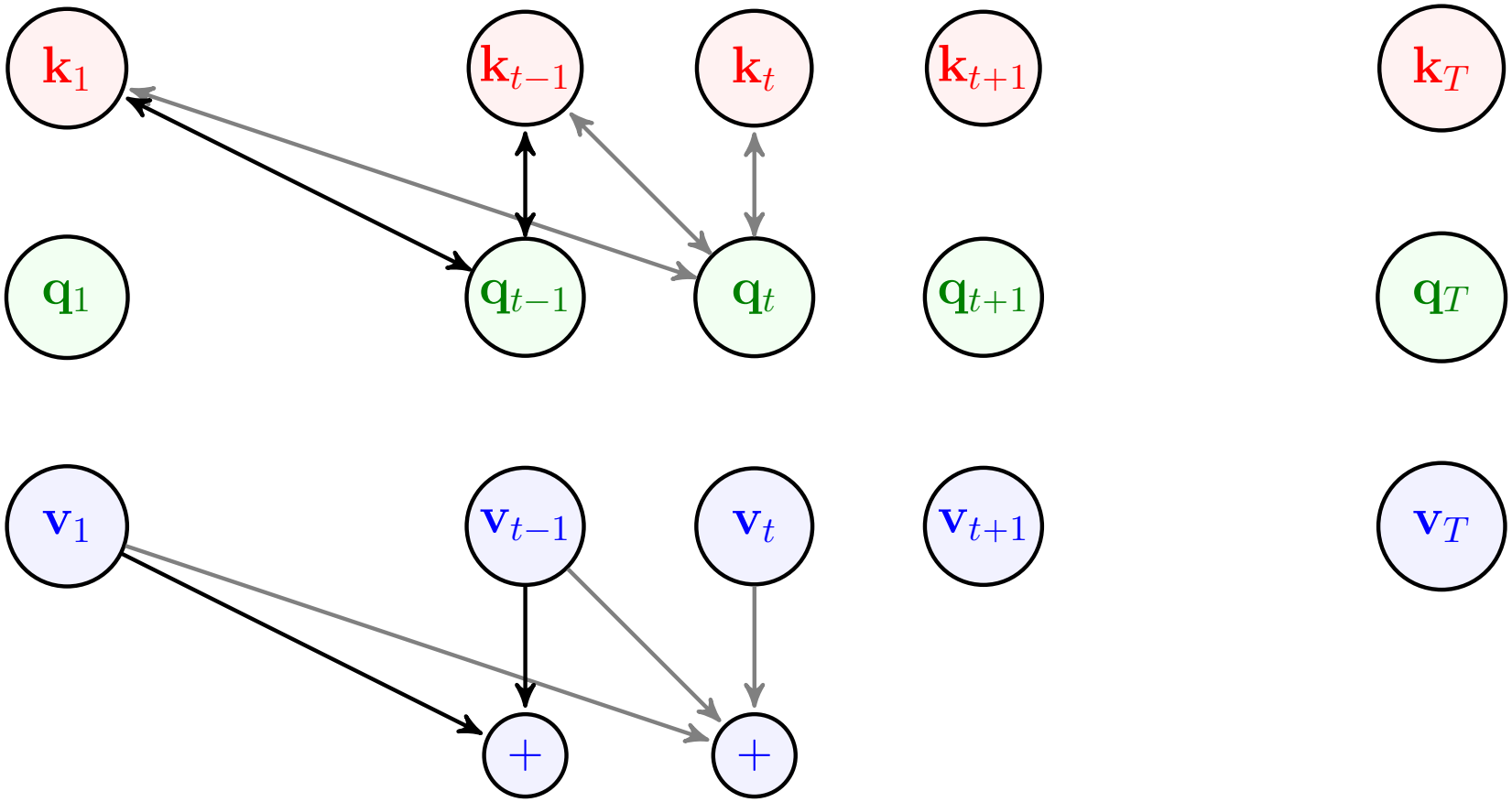


Classical combining approach is to add

$$\boldsymbol{x}_t = \texttt{embd}(x_t) + \texttt{embd}(t)$$

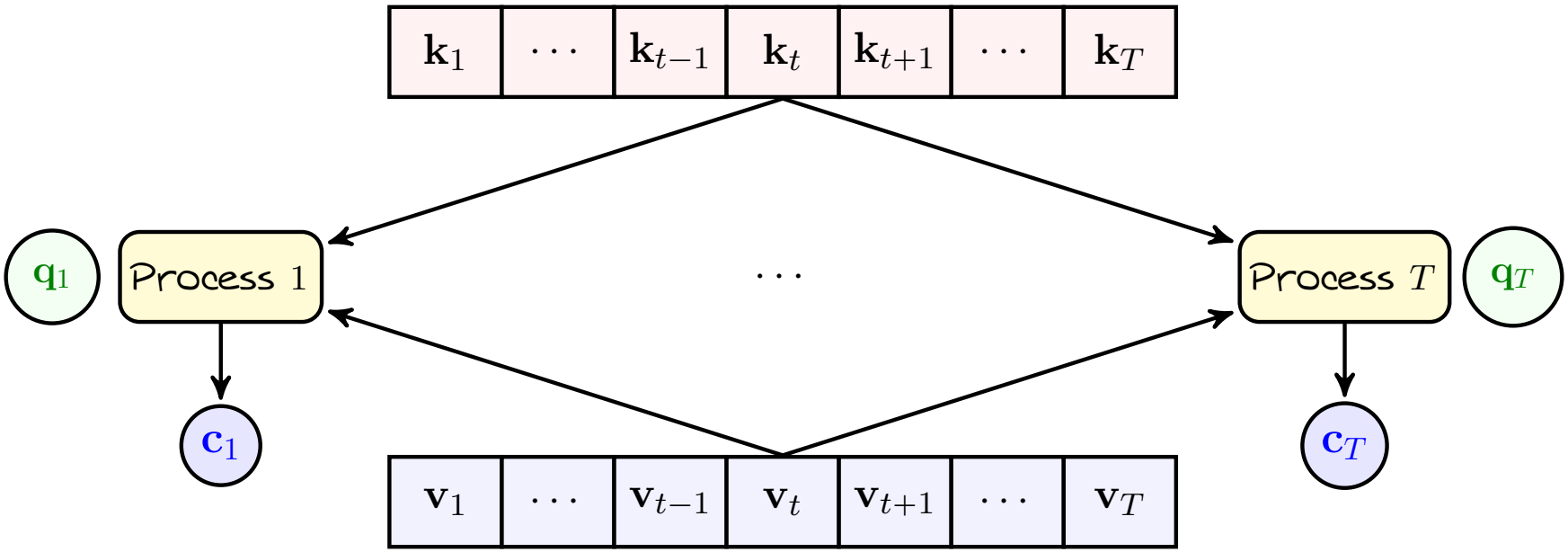
Parallel Context Computation

+ *Do we still need to process one token at a time?*
– No! *The cool part is that we can compute all contexts in parallel*



Parallel Context Computation

Indeed, we can break computation of $\mathbf{c}_1, \ldots, \mathbf{c}_T$ into T parallel processes

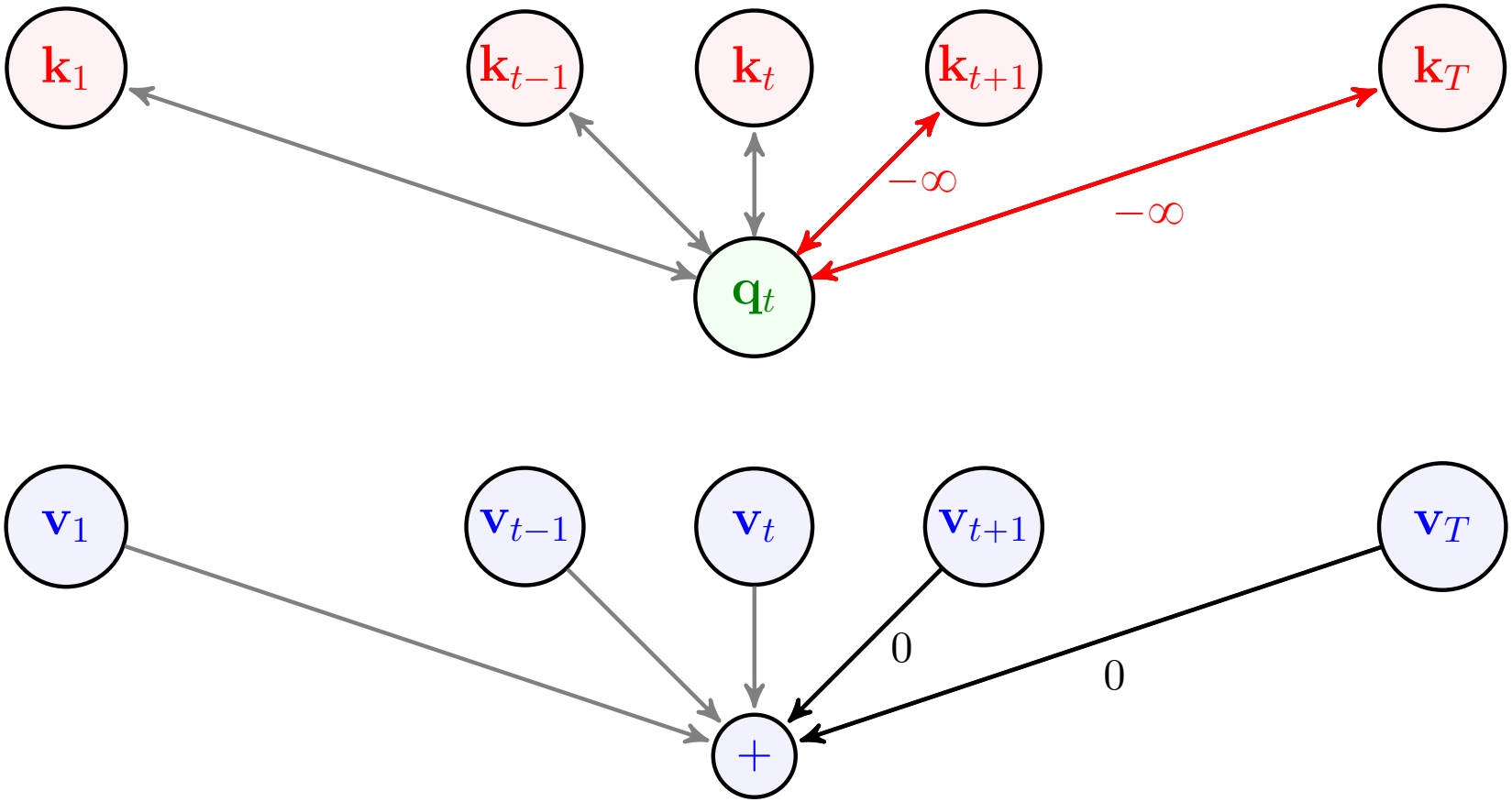


\+ *But the number of weights in each process differ, right? Process 1 computes one coefficient, Process 2 computes two,* $\cdots$

– Yes! And, this can be easily unified ☺

Self-Attention with *Masked Decoding*

We can *get query from all keys in the sequence and rewrite them to keep the weights zero for future tokens* $\rightsquigarrow$ *replace future inner-products with* $-\infty$



Masked Decoding

+ *Why do we replace the inner products with $-\infty$?*
– To make the corresponding Softmax output zero

By setting $\exp\{\xi_{j\to t}\} = -\infty$ for $j > t$, we have

$$\alpha_{j\to t} = \frac{\exp\{\xi_{j\to t}\}}{\sum_{i=1}^{T}\exp\{\xi_{i\to t}\}} = \frac{\exp\{-\infty\}}{\sum_{i=1}^{T}\exp\{\xi_{i\to t}\}} = 0$$

+ *Why do we call it masked decoding?*
– Well! If we wanted *future tokens impact the current one*
  ↳ *we did not need to mask out $\exp\{\xi_{j\to t}\}$ for $j > t$*
  ↳ *this is used encoding, e.g., for translation machine*

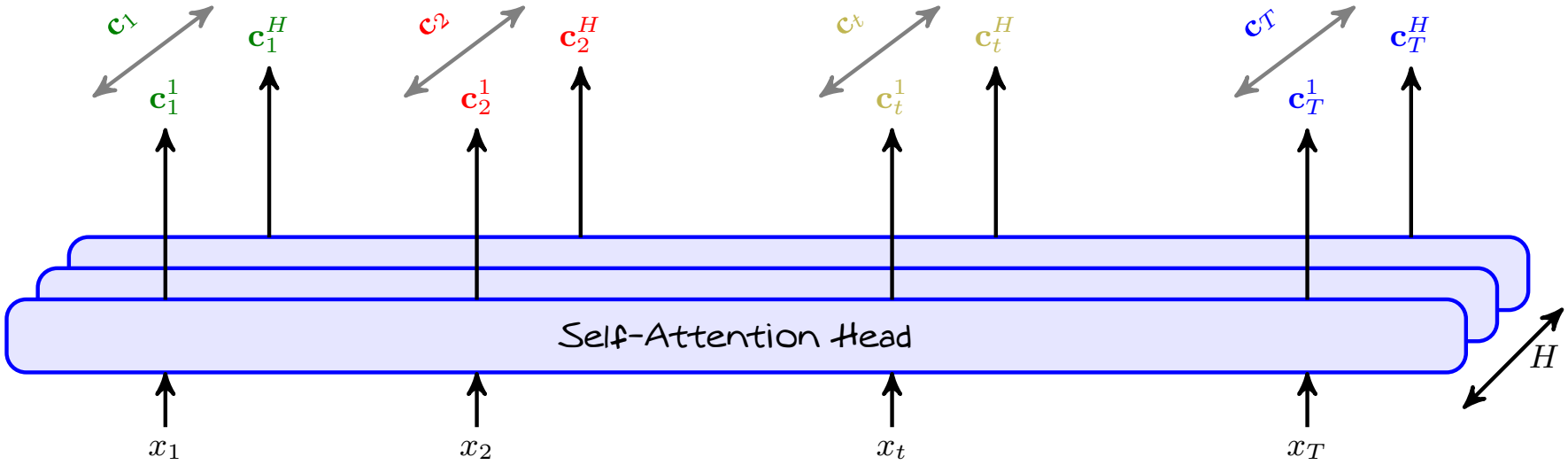
Self-Attention: *Single Head*

SA_Head($x_{1:T}$: input_seq) :

1: **for** $t = 1 : T$ **do**
2: Set $\boldsymbol{x}_t \leftarrow \texttt{embd}\left(x_t\right) + \texttt{embd}\left(t\right)$ #positional enc
3: $\mathbf{k}_t, \mathbf{q}_t, \mathbf{v}_t \leftarrow \texttt{Linear}\left(\boldsymbol{x}_t\right), \texttt{Linear}\left(\boldsymbol{x}_t\right), \texttt{Linear}\left(\boldsymbol{x}_t\right)$
4: $\xi_{1:T \rightarrow t} = \langle \mathbf{k}_1; \mathbf{q}_t \rangle, \ldots, \langle \mathbf{k}_T; \mathbf{q}_t \rangle$ #query $\times$ keys
5: **if** masked_decode = True **then**
6: $\xi_{t+1:T \rightarrow t} = -\infty$
7: **end if**
8: $\alpha_{1:T \rightarrow t} = \mathit{Soft}_{\max}\left(\xi_{1:T \rightarrow t}\right)$ #attention weights
9: $\mathbf{c}_t \leftarrow \sum_i \alpha_{i \rightarrow t} \mathbf{v}_i$ #build context
10: **end for**
11: **return** $\mathbf{c}_t$ for $t = 1 : T$

+ *Why calling it a Head?*
– Well! In practice *we can apply multiple of these heads in parallel to make a richer context*

Self-Attention: *Multi-Head*



Head Size

We typically set the size of context in each head such that the concatenated context is of embedding size, i.e.,

$$\mathbf{c}_t \updownarrow E \rightsquigarrow \mathbf{c}_t^h \updownarrow \frac{E}{H}$$

Self-Attention: *Multi-Head*

`SelfAttention(`$x_{1:T}$`:input_seq, `H`:# of heads):`

1: **for** $h = 1 : H$ **do**
2: Compute $\mathbf{c}_{1:T}^{h} \leftarrow$ `SA_Head(`$x_{1:T}$`)` `#independent heads`
3: **end for**
4: Set $\mathbf{c}_t \leftarrow \left[\mathbf{c}_t^1, \ldots, \mathbf{c}_t^H\right]$
5: **return** $\mathbf{c}_t$ for $t = 1 : T$

\+ *Is this then a Transformer?*

– A shallow one! We can make it deep with L layers!

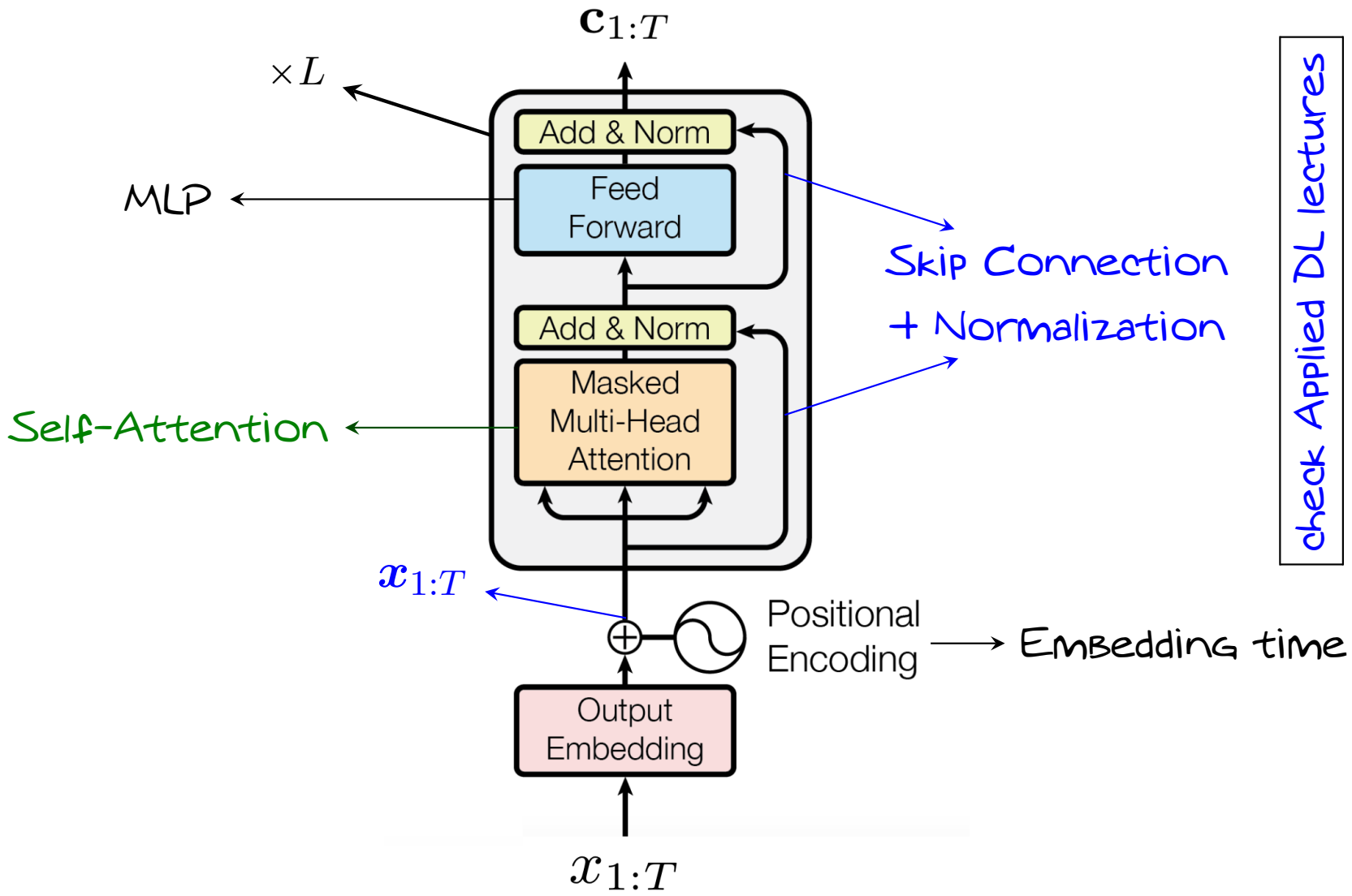
Transformer-Based LM: *Context Extractor*

TransformerDec($x_{1:T}$, H:# of heads, L:depth):

1: **for** $\ell = 1 : L$ **do**
2: $\quad \mathbf{c}_{1:T} \leftarrow \texttt{SelfAttention}(x_{1:T}, H)$
3: $\quad \mathbf{c}_t \leftarrow \texttt{MLP}(\mathbf{c}_t)$ *for* $t = 1 : T$ #extra computation
4: **end for**
5: **return** $\mathbf{c}_t$ for $t = 1 : T$

+ *What does the MLP do?*
- It makes more complexity to the model
  - ↳ *This increases the model capacity*
  - ↳ *It can hence enhance the learning capability of the model*
+ *Good that we talk again "computational"!* ☺
- We need to switch to *"statistical" talks* pretty soon ☺

Transformer-Based LM: *Overview*



Transformer-based LM: *Context Extractor*

+ *Wait a moment! We have not computed the distribution!*
– We can readily do it by a final MLP

`TransformerLM(`$x_{1:T}$`,` H`:# of heads,` L`:depth):`

1: $\mathbf{c}_{1:T} \leftarrow$ `TransformerDec(`$x_{1:T}, H, L$`)`
2: **for** $t = 1 : T$ **do**
3: $\quad \mathbf{p}_{t+1} \leftarrow$ `MLP(`$\mathbf{c}_t$`)` $\qquad \#\mathbf{p}_{t+1} \in [0, 1]^I$
4: **end for**
5: **return** $\mathbf{p}_{t+1}$ for $t = 1 : T$

Training Transformer-based LM

It's good to imagine how we can train this model

`TransformerLM_Train(`$\mathbb{D}$`:train_set):`

for *multiple epochs* **do**
  Sample a batch $\mathbb{B} = \{(x_{1:T+1})_b\}$ `#samples of` T `tokens`
  for $b = 1 : B$ **do**
    $\mathbf{p}_{2:T+1,b} \leftarrow$ `TransformerLM(`$x_{1:T,b}, H, L$`)`
    Compute $\nabla_b \leftarrow -\sum \nabla \log \mathbf{p}_{t+1,b}[x_{t+1,b}]$ `#seq LL grad`
  end for
  Update $\mathbf{w} \leftarrow \mathbf{w} - \eta$`Opt_avg` $\{\nabla_b\}$ `#average batch`
end for

Important

Training loop in this case can be extensively parallelized!

Generation via Transformer-based LMs

It's good to think how a this LM generates the whole new text

`TransformerLM_Generation(`$x_{1:t}$`:input_text):`

1: **for** $i = t : T-1$ **do**
2: $x_{1:T+1} = x_{1:i}, 0, \ldots, 0$ `#add` $\varnothing : 0$
3: $\mathbf{p}_{2:T+1} \leftarrow$ `TransformerLM(`$x_{1:T-1}, H, L$`)` `#masked decode`
4: $x_{i+1} \leftarrow$ `multinomial(`$\mathbf{p}_{i+1}$`, #smpl=1)` `#sample next token`
5: **end for**
6: **return** $\{$`token`$[x_t]$ *for* $t = 1 : T\}$ `#completed text`

In practice though, we can improve computational efficiency by

- *We can drop number of key and query computations by caching*
- *We compute only the target token distribution in each iteration*
- ...

Ready for LLMs!

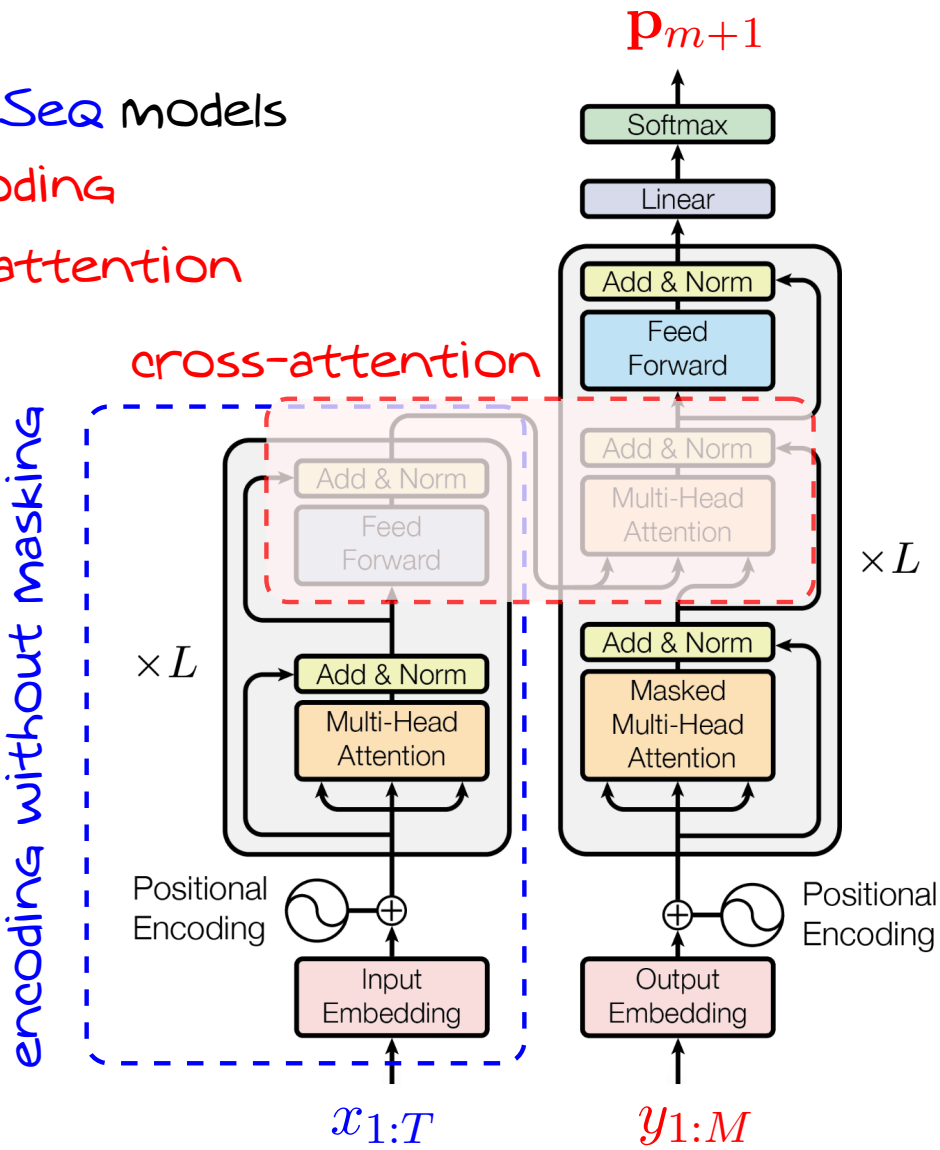
+ *Is this then an LLM?*
– Well! If scale it crazy up, Yes!
+ *How crazy should we scale?*
– Pretty crazy! ☺
+ *But, isn't this simply a text completer? How does it do what ChatGPT does?*
– Well! This is what we call *"Pre-trained LLM"*; we need to do some few extra steps to get there
+ *What are those steps?*
– No worries! We have the next section dedicated to LLMs

Final Note on Transformers

Transformers are Seq2Seq models

↳ our LM uses only decoding

↳ we don't need cross-attention



check Chapter 7 of Applied DL lectures